

Capability is a *system parameter*.

Every complex system depends on what people can do inside it. That dependency is measurable, designable, and too important to leave to chance. LENS is the Johns Hopkins concentration that engineers it.

THE PROBLEM

Systems don't fail because the technology breaks. They fail because *someone couldn't do what the system required*.

Hardware gets reliability analysis. Software gets tested to specification. Human capability gets a training program bolted on at the end — and when the system underperforms, no one can say whether the gap sat in the people, the design, or the organization around both. LENS treats capability as the interface between what a system demands of its people and the impact it delivers — an interface that can be specified, instrumented, measured, and engineered.

THE FIVE COMPETENCIES

- 1 Systems Analysis** *see the whole system*
Trace how human capability and system performance depend on each other.
- 2 Iterative Development** *build, test, refine*
Build through engineering cycles that survive contact with real operations.
- 3 Human-System Collaboration** *work together*
Design partnerships — human-AI among them — that keep people in command.
- 4 Test and Evaluation** *show what works*
Produce decision-grade evidence; tell a learning gap from a system gap.
- 5 Navigating Sociotechnical Constraints** *make it work in the real world*
Fit the design to the constraints that decide whether it survives in the field.

See the whole system. Build, test, refine. Work together. Show what works. Make it work in the real world.

WHERE IT APPLIES

Representative, not exhaustive

A domain is **high-consequence** when the cost of a capability gap is paid by people who did not choose the risk, and the payment is not fully recoverable.

- Healthcare & patient safety
- Education, training & the learning workforce
- Aviation & aerospace
- Defense & national security
- Industry, energy & enterprise systems
- Disaster prevention & recovery
- Algorithms, governance & public systems

THE PROGRAM

DEGREE

M.Ed. in Learning Design & Technology —
LENS concentration, Johns Hopkins
School of Education with the Applied
Physics Laboratory

FORMAT

Online and asynchronous, part-time, built
for working professionals

SCOPE

10 courses · 30 credits

CAPSTONE

An end-to-end project on a real
capability problem at a partner
organization

STANDARDS

Aligned to the emerging IEEE ICICLE
learning engineering competencies

WHO IT'S FOR

Two audiences, equally: education and
training professionals moving toward
systems practice, and engineers and
technical professionals moving toward
learning design.

No one carries every tool of the field.
LENS develops integrators who can lead
and work within a learning engineering
team.

250K+

DEATHS A YEAR, CONTESTED ESTIMATE

attributed to medical error in the U.S. The figure is
disputed; what it counts is not individual mistakes
but system-wide failings and poorly coordinated
care.

Makary & Daniel, BMJ, 2016 · Johns Hopkins

~10 yr

OF UNMET TRAINING REQUIREMENTS

The Marine Corps has been unable to meet
training requirements in the Indo-Pacific — the
theater the National Defense Strategy designates
as the department's top priority.

GAO-24-107463 · 2022 National Defense Strategy

\$100M

SPENT, THEN SHUT DOWN IN 14 MONTHS

inBloom collapsed not because the technology
failed, but because governance, privacy, and
stakeholder trust were treated as afterthoughts
rather than design constraints.

Data & Society, 2017 · nine states withdrew